

With hypothesis testing, we are interested in discovering whether there is evidence in the data for a hypothesis of interest

For example, when a company is interested in studying a new drug, they may seek evidence that the new drug performs better than an existing drug or a control

Hypotheses are typically statements about population parameters

Let $f(x; \theta)$ be a family of distributions, $\theta \in \Theta$

Definition: The two complementary hypothesis in a hypothesis testing problem are called the **null hypothesis** and the **alternative hypothesis**. They are denoted by H_0 and H_1 , respectively:

$$H_0 : \theta \in \Theta_0, \quad H_1 : \theta \in \Theta_1,$$

where Θ_0 and Θ_1 are two disjoint subsets of the parameter space Θ

e.g. You find a coin & want to check if it's fair or not.

H_0 : Coin is fair, $\theta = 1/2$

H_1 : Coin is unfair, $\theta \neq 1/2$

Expt. Toss the coin 100 times & record the # of heads

$$X \sim \text{Bin}(100, \theta)$$

If $\theta = 1/2$, X should be close to ?

Test could be $\text{if } |X - 50| < t, \text{ then } \theta = 1/2$

$\text{if } |X - 50| > t, \text{ then } \theta \neq 1/2$

What is t ? That decides what is a good test vs bad one?

The null hypothesis could be $H_0 : p = \frac{1}{2}$, so that $\Theta_0 = \{\frac{1}{2}\}$

So, the alternative hypothesis is $H_1 : p \neq \frac{1}{2}$, so that $\Theta_1 = \Theta - \Theta_0 = [0, \frac{1}{2}) \cup (\frac{1}{2}, 1]$

This null hypothesis is an example of a **simple hypothesis**, where the parameter can only be a single specified value

Any other hypothesis, including our alternative hypothesis, is a **composite hypothesis**

Generally, the null hypothesis is of the form "No difference", "no change", "status quo", and the alternative as the one researchers are interested in, i.e. "Some effect", "some difference", "some change"

We seek **evidence** in the data for the **alternative hypothesis** – if we find it, we reject the null hypothesis

But we never seek **proof for the null hypothesis** (innocent until proven guilty)

Questions in real life

- 1 Does smoking cause lung cancer?
- 2 Is global warming real?
- 3 Are men more likely to run a stop sign than women?
- 4 Does chemotherapy really cure cancer?
- 5 Is a new drug lower your blood pressure?

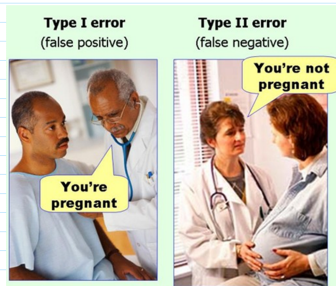
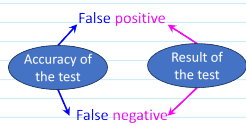
Statistical tests are typically used to make a decision about these real life questions.

Types of Error

- There is a clear analogy between a hypothesis test and a courtroom trial.
- Mistakes can happen if either
 - an innocent person is convicted, or
 - a guilty person is set free (not convicted).
- In statistics, we call these mistakes 'errors' and there are two types of error...

	H_0 True	H_0 False
Reject H_0	Type I Error	
Fail to Reject H_0		Type II Error

*: not an example from real life



People are **generally** not pregnant!

Default	You are not pregnant	H_0
Alternative	You are pregnant	H_1

Justice system in a democracy

- How can someone be arrested if they really are **presumed innocent**?
- Why is a defendant pronounced "**Not Guilty**" instead of innocent?
- Why do citizens put up with a system that allows **criminals to go free on technicalities**?

Common challenges in justice, research (life?)

- A **strong urge** to believe in unusual event/discovery
 - A charged person (aka "**defendant**") is the **convict**
 - This new medicine **cures** Alzheimer's/cancer
 - New concrete mixing yields **higher strength/durability**
- Huge repercussions of wrong decision

You acted on your belief, BUT The belief turned out to be FALSE	You did NOT act on your belief, and the belief turned out to be TRUE

- **No sure-shot way** to prove the unusual event
- **Limited data**

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You acted on your belief, BUT The belief turned out to be FALSE	You did NOT act on your belief, and the belief turned out to be TRUE
Innocent being punished	guilty going free
Patients getting the wrong medicine (side effect/death?)	Patients suffering/dying, despite a discovered medicine
Buildings constructed using inferior material	Wasted resources, unsustainable

- **No sure-shot way** to prove the unusual event
- **Limited data**

Hypothesis testing is the solution!

8

Hypothesis testing (& justice system)

No numerical values in courts, but they share four common features:

- 1. The alternative hypothesis:** This is why a *criminal is arrested*.
 - The police, of course, do not think that the criminal is innocent.
 - The researchers think that their treatment is effective. H_1 or H_A .
- 2. The null hypothesis:** The *presumption of innocence*.
 - The suspect or treatment didn't do anything. H_0 is the logical opposite of H_1 .
- 3. A standard of justice:** A *reasonable doubt*. A test score!
 - No possibility of absolute proof. So, a standard has to be set.
 - Reject the null hypothesis beyond a reasonable doubt.
- 4. A data sample:** Evaluation of *partial information*.
 - Eye-witnesses/fingerprints/DNA analysis/experimental/numerical data of treatment.
 - Getting the "whole truth and nothing but the truth" is often impossible.

Hypothesis testing (& justice system)

Both statistical testing and the justice system:

- Concentrate on **rejecting the null hypothesis**
 - It's much easier
 - Rejection of presumption of innocence $\equiv$ defendant is pronounced guilty
- Consider a failure to reject the null hypothesis
 - As "Not guilty" verdict.
 - "medicine does not treat cancer/concrete is not stronger".
 - Proving H_0 (the null hypothesis of innocence) will take endless evidence.

Hypothesis testing (& justice system)

Neither statistical testing and the justice system are perfect:

- Sometimes, the jury makes an error.
 - An *innocent person goes to jail*
 - Statisticians call it a
- Sometimes, a *guilty person is set free*
 - Statisticians call it a

Which one is worse?

Errors in Hypothesis testing (& justice system)

$$\text{Type I error} \equiv \left\{ \begin{array}{l} \text{an innocent person goes to jail} \\ + \\ \text{a guilty person walks free} \end{array} \right\}$$

HT (& justice system) puts a lot of emphasis on avoiding Type I error.

Counter example: Airport security

Innocent person and alarm goes off (Type I error)

Guilty person and alarm goes not go off (Type II error)

Sometimes, Type II error is worse!

Spot the error type

According to a report from the United States Environmental Protection Agency, burning one gallon of gasoline typically emits about 8.9 kg of CO₂. A fuel company wants to test a new type of gasoline designed to have lower CO₂ emissions. Here are their hypotheses:

$$H_0: \mu = 8.9 \text{ kg}$$

$$H_a: \mu < 8.9 \text{ kg}$$

$$H_0: \mu = 8.9$$

$$H_1: \mu < 8.9$$

(where μ is the mean amount of CO₂ emitted by burning one gallon of this new gasoline).

Under which of the following conditions would the company commit a Type I error?

Choose 1 answer:

- (A) The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- (B) The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they conclude it is lower than 8.9 kg.
- (C) The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they conclude it is lower than 8.9 kg. *Reject H₀*
- (D) The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they fail to conclude it is lower than 8.9 kg.

A quality control expert wants to test the null hypothesis that a new solar panel is no more effective than the older model.

$$H_0: \mu_{\text{new}} = \mu_{\text{old}}$$

$$H_1: \mu_{\text{new}} > \mu_{\text{old}}$$

What would be the consequence of a Type I error in this context?

Choose 1 answer:

- (A) They don't conclude the new panel is more effective when it actually is not more effective.
- (B) They don't conclude the new panel is more effective when it actually is more effective.
- (C) They conclude the new panel is more effective when it actually is more effective.
- (D) They conclude the new panel is more effective when it actually is not more effective.

Significance level of the test

$$\alpha = P(\text{Type I error})$$

$$\alpha = P(\text{Reject } H_0 | H_0 \text{ is true})$$

$$\beta = P(\text{Type II error})$$

$$\beta = P(\text{Do not reject } H_0 | H_0 \text{ is false})$$

$$\beta = P(\text{Do not reject } H_0 | H_1 \text{ is true})$$

Typically, we minimize α

coin toss example: How to find α ?

Fix $\alpha = 0.05$

Assume H_0 is true, then $X \sim \text{Bin}(100, 1/2)$

$$Y = \frac{X - 100 \cdot \frac{1}{2}}{\sqrt{100 \cdot \frac{1}{2} \cdot (1 - \frac{1}{2})}} \sim Z \sim N(0, 1)$$

$$\alpha = P(X - 50 > c, \theta_0 = 1/2)$$

$$= P(Z > c, \theta_0 = 1/2)$$

CI $\equiv$ Acceptance region
complement of α

A large university is curious if they should build another cafeteria. They plan to survey a sample of their students to see if there is strong evidence that the proportion interested in a meal plan is higher than 40%, in which case they will consider building a new cafeteria.

Let p represent the proportion of students interested in a meal plan. Here are the hypotheses they'll use:

$$H_0: p \leq 0.40$$

$$H_a: p > 0.40$$

$$H_0: p \leq 0.4 \approx \text{not building}$$

$$H_1: p > 0.4 = \text{building}$$

What would be the consequence of a Type II error in this context?

Choose 1 answer:

- (A) They don't consider building a new cafeteria when they should.
- (B) They don't consider building a new cafeteria when they shouldn't.
- (C) They consider building a new cafeteria when they shouldn't.
- (D) They consider building a new cafeteria when they should.

According to a report from the United States Environmental Protection Agency, burning one gallon of gasoline typically emits about 8.9 kg of CO₂. A fuel company wants to test a new type of gasoline designed to have lower CO₂ emissions. Here are their hypotheses:

$$H_0: \mu = 8.9 \text{ kg}$$

$$H_a: \mu < 8.9 \text{ kg}$$

(where μ is the mean amount of CO₂ emitted by burning one gallon of this new gasoline).

Which of the following would be a Type II error in this setting?

Choose 1 answer:

- (A) The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they conclude it is lower than 8.9 kg.
- (B) The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- (C) The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- (D) The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they conclude it is lower than 8.9 kg.

$$Y = \frac{\bar{X} - 100 \pm \frac{1}{2}}{\sqrt{100 \times \frac{1}{2} \times (1 - \frac{1}{2})}} \sim \mathcal{N}(0,1)$$

$$\alpha = P(|X - 50| > t, \theta_0 = \frac{1}{2})$$

$$= P(|Z| > \frac{t}{\frac{1}{2}}, \theta_0 = \frac{1}{2})$$

$$0.05 = P(|Z| > \frac{t}{\frac{1}{2}}) = 2 \cdot 2 \cdot \Phi(\frac{t}{\frac{1}{2}}) - 1 \quad \text{--- (1)}$$

choose t s.t. the eqn (1) is satisfied

$$t = 9.8$$

CI

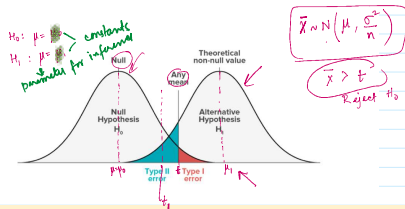
$$\Rightarrow (10.2, 59.8) \text{ leads to } \rightarrow$$

do not reject H_0 (not sufficient evidence)

also reject H_0

CI $\equiv$ Acceptance region
complement of (Rejection Region)
 H_0

Can we minimize both types of errors simultaneously?

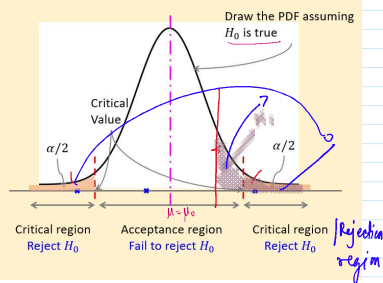


$$H_0: \mu = 500$$

$$H_1: \mu \neq 500$$

Checklist:

1. How to draw PDF
2. Critical value
3. Acceptance region
4. Critical region
5. Sig. level
6. Areas
7. When reject/fail to reject H_0



Probability of Type I Error, α

$$\alpha = \Pr(\text{Type I error})$$

$$= \Pr(\text{reject } H_0 \text{ when } H_0 \text{ is true})$$

aka the **significance level** or the α -error

We can reduce α by

- by widening the acceptance region
- by increasing the sample size n

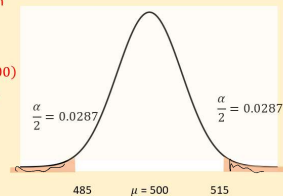
Probability of Type I Error, α

- Consider testing $\mu = 500$ vs. $\mu \neq 500$ again; suppose $\sigma = 25$
- Suppose we tested a sample with $n = 10$ specimens
- If a client accepts products with average $f \in [485, 515]$
- Find the **probability of Type I error**

- Type I error will occur when $\bar{x} > 515$ or $\bar{x} < 485$ for $\mu = 500$.

- CLT $\Rightarrow \bar{x}$ is approx. normal with $\mu = 500$ and std = $\sigma/\sqrt{n} = 7.9$

- $\alpha = P(\bar{X} < 485 \text{ when } \mu = 500)$
+ $P(\bar{X} > 515 \text{ when } \mu = 500)$



$$Z(\bar{X} = 485) = -1.90$$

$$Z(\bar{X} = 515) = 1.90$$

$$\alpha = P(Z < -1.90) + P(Z > 1.90)$$

$$= 0.0287 + 0.0287 = 0.0574$$

$$P(\bar{x} < 485 \text{ or } \bar{x} > 515 | H_0) \mid \bar{x} \sim N(\mu, \frac{\sigma^2}{n} = \frac{(25)^2}{10})$$

$$= P(\bar{x} < 485 | H_0) + P(\bar{x} > 515 | H_0)$$

Probability of Type II Error, β

• $\beta = \text{Pr}(\text{Type II error})$

= $\text{Pr}(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$
aka the β -error

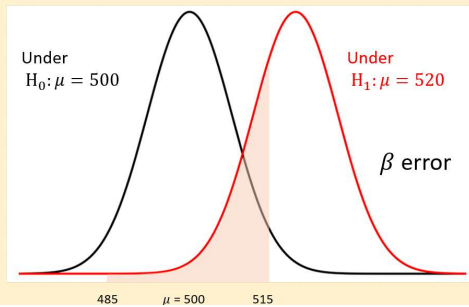
• To calculate β we must have a **specific alternative hypothesis**,

For example: choose a specific alternative value, e.g., 520,

$$H_0: \mu = 500 \text{ versus } H_1: \mu = 520$$

Type II error

Type II error will be committed if $\bar{x} \in [485, 515]$ when $\mu = 520$
 $\beta = P(485 \leq \bar{x} \leq 515 \text{ when } \mu = 520)$



• Find the **probability of Type II error?**

• A type II error will occur when $\bar{x} \in [485, 515]$ for $\mu = 520$ MPa.

• $CLT \Rightarrow \bar{x} \sim N(\mu = 500; \text{std} = 7.9)$ (under H_0)

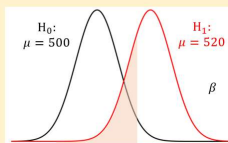
• $\beta = P(\bar{X} \in [485, 515] \text{ when } \mu = 520)$

$Z(\bar{X} = 485) = -4.43$

$Z(\bar{X} = 515) = -0.63$

• $\beta = \text{Pr}(Z \leq -0.63) - \text{Pr}(Z \leq -4.43)$

= $0.2643 - 0.0000 = 0.2643$



Power of a statistical test

• The **power** of a statistical test

$\text{Pr}(\text{rejecting } H_0 \text{ when } H_1 \text{ is true})$

$$\text{Power} = 1 - \beta$$

interpreted as the **probability of correctly rejecting a false null hypothesis**

Important Properties of a Test of Hypothesis

1. The type I error and type II error are related. A decrease in the probability of one generally results in an increase in the probability of the other.
2. The size of the critical region, and therefore the probability of committing a type I error, can always be reduced by adjusting the critical value(s).
3. An increase in the sample size n will reduce α and β simultaneously.
4. If the null hypothesis is false, β is a maximum when the true value of a parameter approaches the hypothesized value. The greater the distance between the true value and the hypothesized value, the smaller β will be.

p-value

= $P(\text{observing as extreme, obs. as you have observed} | H_0)$
 $H_0: \mu = \mu_0$

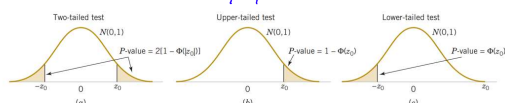


FIGURE 9-10 The P-value for a z-test. (a) The two-sided alternative $H_1: \mu \neq \mu_0$. (b) The one-sided alternative $H_1: \mu > \mu_0$. (c) The one-sided alternative $H_1: \mu < \mu_0$.

Whether one sets up a one-tailed or a two-tailed test will depend on the conclusion to be drawn if H_0 is rejected.

For example, in testing a new drug, one sets up the hypothesis that it is no better than similar drugs now on the market and tests this against the alternative hypothesis that the new drug is superior.

However, if we wish to compare a new teaching technique with the conventional classroom procedure, the alternative hypothesis should allow for the new approach to be either inferior or superior to the conventional procedure.

Example 10.1: A manufacturer of a certain brand of rice cereal claims that the average saturated fat content does not exceed 1.5 grams per serving. State the null and alternative hypotheses to be used in testing this claim and determine where the critical region is located.

Example 10.2: A real estate agent claims that 60% of all private residences being built today are 3-bedroom homes. To test this claim, a large sample of new residences is inspected; the proportion of these homes with 3 bedrooms is recorded and used as the test statistic. State the null and alternative hypotheses to be used in this test and determine the location of the critical region.

Approach to Hypothesis Testing with Fixed Probability of Type I Error	1. State the null and alternative hypotheses. 2. Choose a fixed significance level α. 3. Choose an appropriate test statistic and establish the critical region based on α. 4. Reject H_0 if the computed test statistic is in the critical region. Otherwise, do not reject. 5. Draw scientific or engineering conclusions.
Significance Testing (P -Value Approach)	1. State null and alternative hypotheses. 2. Choose an appropriate test statistic. 3. Compute the P-value based on the computed value of the test statistic. 4. Use judgment based on the P-value and knowledge of the scientific system.